
Cerebro

Operator-Owned AI Infrastructure as a Governance Discipline

From Architecture Thesis to Evidence-Bounded Capability

Cerebro Cyber Solutions

Version v2.2

Date 14 August 2026

Evidence current as of 14 August 2026.

Internally exercised infrastructure. Claim-bounded evidence. External validation still to be earned.

Document control

Field	Value
Document	Cerebro White Paper v2.2
Version	2.2
Evidence freeze	Merged main as of 14 August 2026; exact commit pin recorded in the repository review notes.
Continuation chain	v1.3.3-public + Supplement 1 -> v2.0 -> v2.1 -> this v2.2 release
Purpose	Refresh the public architecture against material merged evidence through the freeze while separating source-built, internally proven, operating and externally validated state.
Public-claim rule	Repository evidence is dated. Live runtime claims require a fresh owning probe.

STATUS VOCABULARY / Built means merged code or a recorded artifact. Internally proven means a bounded test or dated receipt passed. Operating internally requires an observed owning runtime path. Externally validated requires independent customer or market evidence. Operator-gated, Proposed and Future remain distinct. These labels are not interchangeable.

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READER PATH

For the shortest decision path, read Sections 1, 6, 9 and 14. For architecture depth, add Sections 4, 7, 8 and 10.

1. Executive abstract

Cerebro Cyber Solutions is building a general-purpose AI operations platform as intellectual property. What exists today, and what this paper audits, is a real, internally exercised architecture and verification estate, not yet a repeatable, externally validated platform, in which governance, verification and provenance are properties of the architecture rather than a compliance layer added afterward.

That estate has been exercised across operations, products, business development and content through roughly four months of versioned repository activity, from the first tracked commit on 18 April 2026 to this evidence cut on 14 August 2026. Its differentiator is not a model or an agent runtime but the discipline binding them: the company itself is implemented as inspectable, versioned infrastructure, with auditability, reversibility and human accountability built in rather than documented afterward.

This is an internally exercised repository and verification estate, not merely a diagram. It records ten adopted engineering principles anchored by Verified Completion - a worker cannot grade its own work, an independent verifier must run, and its result must control the material transition - now governed by a cost function that admits new verification only where it protects a real bid, model, artifact or operator capability. Its federated memory spans eight designed collections with a dated internal recall result and degradation that fails visibly rather than silently. Its model factory has trained Cerebro's own models on owned hardware: a cybersecurity finding-triage specialist is promoted and serving, a general-agent student passed a bounded internal comparison and is retained content-addressed and bound to no consumer, and transfer-gate promotion verdicts are graded by an independent, digest-pinned judge model rather than by the worker that produced the candidate. Its receipt ledgers are hash-chained under an honest-operator threat model, so an edit to the ledger is caught rather than trusted. Every figure in this paper is an internal repository snapshot, bounded test result or operating result at the recorded evidence pin, not an external benchmark.

The boundary is stated plainly and held at full force throughout this paper: Cerebro is a real, internally exercised architecture and verification estate - not yet a repeatable externally validated platform. The installable slice, Libro, is a public v0.2.1-alpha pre-release; BlackBox is pre-ship; and the repository holds no claim-grade evidence of a repeatable customer deployment, an externally validated outcome or recurring revenue. Version 2.2 makes that architecture-ahead-of-market gap explicit rather than converting internal activity into external traction.

V2.2 VERDICT / Cerebro is a real, internally exercised architecture and verification estate, but not yet a repeatable externally validated platform. Its strongest evidence is repository assurance, operational memory, one proven specialist model lane and one observed Telegram operator seam. Its weakest evidence is end-to-end execution, adversarial containment, recovery drills and customer outcomes.

2. What changed since v1.3.3

Version 2.2 is a new evidence cut over Version 2.1, not a rewrite of the provenance chain. Prior releases remain in place. This edition is pinned to merged main as of 14 August

2026, with the exact pin recorded in the repository review notes, and excludes unmerged branches, open donor work and runtime facts that lack a dated owning probe. The edition was refreshed on 14 August 2026 to a new freeze over the same claim boundary; the refreshed interval contains security hardening and seam completeness work, not new capability lanes, and is absorbed as such.

Surface	v1.3.3 position	v2.2 evidence position
Governance	Nine active principles	Ten adopted principles; Verified Completion is risk-tiered, and new verification must name the product asset it protects
Architecture	Four operating layers	Four internal layers retained; source-built architecture is broader than the one observed operating seam
Memory	Per-source write + unified-read thesis	File/local federation plus four default Qdrant collections, operational kanban history, bounded reranking and fail-visible vector repair
Agents	Multi-surface routing and early Constellation	Named source-built positions; Composer Telegram seam observed, broader modules code-present/not-running or parked
Verification	Audits, lints and provenance	Fourteen registered lanes: nine blocking, five advisory; branch/path controls, typed continuity and shared serialized receipts
Model production	Roadmap-level capability	One proven operating cyber-triage lane; one text-only Layer-0 artifact internally measured and stored but not consumer-bound
Interface	Command Center direction	Local Command Center and the model factory built; interface presence does not imply general executor operation
Security	Least privilege, origin and scope contracts	Accidental-exposure, path and provenance controls; no claim of adversarial-model containment or reliable namespaced-origin lockout
Recovery	Reversibility as principle	SQLite/database-dump and separate Qdrant capture evidence; service restores, credential rebinding and timed multi-service recovery unproven
Business	Pre-release product posture	Libro remains public alpha; BlackBox pre-ship; no claim-grade repeatable deployment, external outcome or recurring-revenue proof

The cut deliberately distinguishes code presence, internal proof, observed operation and external validation. A source module can be Built while not running. A model tag can be internally proven while unbound to any consumer. A backup can be captured and hash-verified while restore remains untested. These distinctions are now first-class publication rules rather than footnotes.

The documentation and operating architecture also gained stronger conservation controls. Load-bearing changes on trunk are blocked unless they use a feature branch or a logged override. The close-to-open seam now generates its carrier manifest from source, uses one close-epoch selector and seals a hash-bound last-close pointer instead of relying on hand transcription. Review receipts use a shared per-repository sink with serialized appends. Eleven Cerebro-authored Claude runtime skills resolve to one canonical source, eliminating copy drift while making the primary checkout itself a deployment surface. Since the 5 August cut, the seam mechanism itself was completed: the last-close pointer seal is now owned by the pre-commit gate rather than by ritual discipline, the close manifest emits deduplicated coverage anchors so carry-completeness is measurable, and the invoker lint that keeps every fence bound to a live executor now recognizes templated executors.

3. The problem and the thesis

The scarcest resource in high-accountability work is not intelligence; it is trust. Organizations whose decisions must survive outside scrutiny extend trust to people through vetting, to systems through inspection, and to records through evidence. Today's AI inverts that: fluent output, thin evidence. Adoption stalls not because models are weak, but because nothing in the usual stack can show an independent party what was done, by what, under which authority, with which checks.

The original problem remains. A small operator serving regulated or security-sensitive buyers needs speed, auditability, ownership, reversibility, cost control and human accountability at the same time. Commercial agent products usually optimize for convenience and hide the runtime. Research prototypes optimize for capability and defer operations. Traditional compliance programs document systems after they exist. A bootstrapped small business cannot afford those separations.

Cerebro therefore treats the company as an engineered system. Decisions, session records, operating policies, skills, route contracts, state ledgers, model manifests, eval inputs and recovery procedures live in versioned artifacts. Mutable operational state is separated from durable doctrine. Department execution is separated from company governance. A human relationship boundary remains explicit even when drafting, analysis and monitoring are automated.

Tools are commodity, and Cerebro absorbs current ones deliberately, under a governed intake ladder, rather than freezing on a stack. The durable asset is the governed relationship among memory, decisions, models, workflows, evidence and operator authority.

The thesis follows: trust in AI is an architecture problem, not a persuasion problem. It is earned the way trust is earned everywhere else, through evidence an independent party can inspect. That is why selected receipt ledgers are hash-chained under an honest-operator threat

model; why the ship rule requires an independent verifier's result before a material transition; and why this paper separates code-present, internally proven, operating and externally validated state. The evidence layer is not paperwork attached to the product; it is the core of the product.

The platform is deliberately agnostic about its parts. Models, tools and agent runtimes are replaceable; Cerebro is the governed wiring beneath them, the way a prefabricated structure carries its wiring, plumbing and inspection points before any occupant moves in. The design goal is that an organization adopts the governed shell and keeps its own choice of parts, present and future. Today that shell is exercised internally and its installable slice is an alpha; the distance from there to a repeatable external product is stated plainly in this paper.

The durable pattern in enterprise systems is that platforms win by running the business problem, with machine intelligence solving the problems the platform surfaces, not the reverse. Cerebro follows that shape. The object of the system is trustworthy operations: verifiable work, reliable delegation, recoverable failure, efficient use of operator time. The AI inside it is workforce, never the point.

Version 2.2 adds a third clause to the thesis: verification has a cost. Infrastructure is governed when a red verdict can stop a material transition, but adding another verifier is justified only when it protects a real product, model, bid or live operator capability. The goal is not maximal ceremony. It is sufficient evidence at the transition where a false green would matter.

This distinction matters most for a solo operator. There is no large team to absorb a false completion claim, a broken backup or a subtly regressed model. Independent critics and deterministic gates are not bureaucracy added to velocity. They are how velocity stays usable when one person is moving across infrastructure, product, proposals and content.

4. What the platform is designed to do: one governed stack

As a design model, Cerebro describes a five-layer operator experience - memory, reasoning, execution, automation and interface - intended to run as one governed stack, with an internal four-layer placement architecture answering where each capability physically lives. The observed external seam today is a single disclosed executor; internal evidence does not yet establish repeatable customer deployment or end-to-end operation across all five layers. Presenting both views is deliberate: intended capability for the customer, placement for the operator, with governance binding the two.

4.1 The customer-facing five-layer stack

INTERFACE	Chat, Telegram, Command Center, documents and operator controls
AUTOMATION	Skills, agents, schedules, hooks, event triggers and governed handoffs
EXECUTION	Owned local models, specialist seats, tools and deterministic services
REASONING	Tiered local and cloud judgment under consent, routing and cost policy
MEMORY	File identity, human-readable knowledge and federated warehouse recall
OPERATOR + GOVERNANCE + VERIFIED COMPLETION span every layer	

Figure 1. The intended customer-facing five-layer design frame, governed as one stack.

In this design model, memory would hold the customer's retained knowledge and operational history. Reasoning is judgment capacity selected by policy. Execution is the local or bounded workhorse capability that performs tasks. Automation composes those capabilities into repeatable flows. Interface is the human trust surface: documents, chat, Telegram, dashboards, the Command Center and explicit approval points. The operator and the governance discipline cut across all five layers.

4.2 The internal four-layer operating architecture

Operating layer	Primary responsibility	Governance posture
Brain	Decisions, sessions, doctrine, evidence and department context	Append-mostly; supersede rather than erase
Skills	Reusable workflow contracts and execution rituals	Declared scope, lint, tests and promotion gates
Agent runtime	Composer, operator routes, specialist seats, hooks and daemons	Role isolation, allowlists, separate checker
Fleet	Compute, model routing, memory services and private networking	Origin policy, placement policy, probe-qualified state

Brain owns why: decisions, sessions, evidence and the chain of supersession. Skills own repeatable how: contracts, scripts, rituals and tests. Agent Runtime owns who or what executes: Composer, specialist seats, routing logic, hooks and scheduled workers. Fleet owns where computation and state run: services, model gateways, stores, workstations and GPU capacity. The Master Brain governs cross-company placement; department brains own lane-specific execution detail.

4.3 The control plane that binds both views

The control plane is not a fifth operating layer. It is the set of cross-cutting constraints that make the layers behave as one system: data classification, route policy, cost caps, model origin, scope contracts, evidence manifests, health probes, promotion gates and human sign-off. This is why the architecture is better described as infrastructure-as-IP than as an agent bundle. The value lies in the relationships and enforcement seams.

ALIGNMENT RULE: Customer language may simplify the stack, but it may not erase internal boundaries. A sales-facing 'memory layer' still decomposes into index, harness and orchestration responsibilities; a polished interface never upgrades a red or unproven backend.

5. The ten-principle governance discipline

The principle set is the stable governance vocabulary. The first nine constrain architecture and operating behavior. Principle #10 determines whether evidence is strong enough to declare completion. The framework is not a values poster; each principle points to a class of enforcement or a required artifact.

#	Principle	Constraint in practice
1	Governance	Company-wide doctrine stays in Master Brain; department execution stays in the owning brain
2	Parity	Skills, docs and enforcement move with the architecture they represent
3	Lockstep	Rollups and state views update with the source artifact, not weeks later
4	Human-in-the-Mix	Relationships, external commitments, representations and strategic sign-offs remain human-owned
5	Reversibility	Supersede rather than erase; retain rollback, receipts and prior state
6	Observability	Silent failure is a defect; long work, corrections and degradations emit evidence
7	Least-Privilege	Tools, paths, network, credentials and write scope are explicit and narrow
8	Reproducibility	Inputs, versions, hashes, routes and stage decisions are replayable
9	Self-Sufficiency	Cerebro owns the control plane; one named operator-messaging executor seat is a permanent disclosed pinned component
10	Verified Completion	Independent evidence decides done; the critic can stop the ship condition

The principle set also explains why the repository contains apparently redundant artifacts. A replaced decision remains because Reversibility requires the supersession chain. A generated dashboard remains distinct from a decision because Governance distinguishes view from authority. A model adapter can remain on disk after rejection because a retained experiment is not the same thing as a served model.

Two doctrine corrections matter in this edition. Self-Sufficiency #9 now carries one named exception: the operator-messaging executor seat is a permanent, disclosed and pinned open-source component, while Cerebro owns the surrounding seam and control plane. Separately, the verification-work admission rule makes product protection the default test for new guard work. Neither correction creates a new product capability; both make the operating truth more defensible.

6. Verified completion: from audit report to ship gate

Principle #10 emerged from two repeated failure shapes. In one, a demo could print a safe narrative while the underlying policy test was red. In the other, a loop could stop when its own worker emitted a promise string. In both cases the worker effectively graded its own homework. The fix is structural: the maker cannot control the final checker, the checker must run, and the checker's result must control the transition.

Leg	Purpose	Failure it prevents
Deterministic test	Assert behavior that can be decided exactly	A fluent narrative hiding a mechanical defect
Non-deterministic eval	Measure model or agent behavior over a bounded rubric or case set	One successful demo being treated as reliable performance
Separate-checker critic	Give an independent system permission to say no	Maker equals checker
Trajectory check	Prove the required verification step actually ran	A skipped test, or a verifier that no process invokes, being reported as completion

Blocking enforcement is risk-tiered. Money, autonomous and promotion lanes receive hard-fail treatment when their contract requires it. Lower-risk or environment-dependent lanes can remain advisory, but advisory is not promotion. At this evidence freeze the registry carries fourteen lanes: nine blocking and five advisory. The registry itself states that this is a trusted-worker guardrail, bypassable by logged override or direct repository mutation, not a tamper-proof security boundary.

The verification spine now includes load-bearing path/branch enforcement, generated close manifests with a sealed last-close pointer, rendered-history trust-domain separation, passing-receipt-only reconciliation, per-repository tamper-evident receipt sinks and serialized ledger appends. An August completeness pass extended the same spine: hook documentation is regenerated from the live registry with a registry-to-configuration lockstep check, and a branch-security-surface-drift detector emits red or review verdicts when a feature branch's tracked security surfaces drift from main. The detector is wired advisory - detection, not commit-blocking, with blocking enforcement an operator-gated decision still open - and a startup-file environment bypass found during its own adversarial review was sealed within that advisory scope. These controls strengthen repository assurance. They do not prove customer outcomes, universal independent review or adversarial containment; scoped acknowledgements and operator escapes remain part of the model. A client-side merge gate now sequences trunk merges against the authoritative remote test run rather than racing it, holding each merge until the slowest verification check concludes green and refusing it on red.

CORE RULE: A report can inform. A gate can govern. Cerebro's v2 direction is to make the verifier's verdict the transition condition wherever a false green would create material risk.

7. Memory as a federated, failure-tolerant system

Memory remains one of Cerebro's strongest architectural differentiators because it is treated as a system of distinct responsibilities rather than a single database. Identity memory, human-readable reasoning, semantic warehouse recall, process state, chronological audit records and active work have different lifecycles. Collapsing them would make the system simpler to describe but harder to govern.

7.1 Per-source write, federated read

Each writer writes the store it owns. Session workflows write decisions and sessions. File-memory tools write per-entry memories. Runtime services write their own state. A routed warehouse path carries decisions, sessions, the memory stream and operational kanban history. Retrieval, not writing, performs unification. Signed admission can bind approved warehouse payloads, but a valid receipt proves writer and integrity properties, not semantic safety or relevance.

The recall corpus is designed as eight collection classes: decisions, sessions, skills, knowledge, the memory stream, research, the operator messaging stream and operational kanban history. The designed classes are stable; how many collections a live instance holds at any moment is a probe-owned fact, not a number this paper fixes.

Tier or store	Primary job	Read posture
Identity file memory	Always-loaded rules, preferences and current behavioral constraints	Local and fast; three explicit surface-specific stores are unioned
Reasoning knowledge	Human-readable associative notes and structured knowledge	Retrieved through owning knowledge surfaces
Qdrant warehouse	Full-history semantic recall across routed collections	Optional federated lane over decisions, sessions, memory stream and operational kanban by default; auth-enforced and failure-tolerant
Process databases	Agent/runtime working state	Not silently represented as human memory recall
Decisions and sessions	Chronology, authority and audit replay	Included through local corpus and warehouse paths

7.2 Retrieval lanes and graceful degradation

The always-on recall path scores a local corpus with lexical relevance, path signal and corpus weighting. Optional lanes add vector similarity, four default Qdrant collections and final cross-encoder reranking; reciprocal-rank fusion combines available lanes. A local copy wins a filename collision. If a warehouse, embed route or GPU reranker is unavailable, retrieval preserves the pre-enhancement result and reports the degraded lane rather than failing the request.

The dedicated reranker remains reached through the model gateway with a route-scoped consumer credential. The July card-recall investigation showed why end-to-end inputs matter: widening the candidate pool alone did not repair the miss, and richer cross-encoder text alone did not repair it; together they moved the target from rank 41 to 6. The change was admitted only with a non-regression control on the broader golden set.

7.3 The July audit and the meaning of honest health

The July audit found two forms of silent blindness. The original coverage metric compared unrelated totals, and the discovery path silently capped the corpus before later sources could enter. Subsequent repairs replaced aggregate arithmetic with exact-path evidence, expanded the rotating golden set, added operational history, bounded rerank work and validated cached vectors for dimension and numerical safety. Oversized documents now degrade visibly instead of being cached as permanent null results.

A dated 3 August real-surface run reported recall@5 of 0.957, vector embedding healthy, reranking active and a shorter completion time after the oversize fix. The earlier four-query card-shaped set remained only partially proven until the wider-input change; the broader set and the card-specific set are not interchangeable. Health remains probe-qualified, and no live backfill claim is made without snapshot and restore evidence.

One consequence of this architecture is easy to miss: the memory system is the estate's own historian. The dated decisions, session records and receipts behind the models, gates and seats described in this paper were captured as the work happened, and much of that history is retrievable through the same recall path the operator uses for daily work. Not every bench artifact or run receipt is indexed for recall; those remain repository files reached by direct review. This paper is assembled from those records, not reconstructed from recollection.

8. Runtime sovereignty, specialist seats and fleet placement

Self-Sufficiency no longer means every load-bearing runtime must be Cerebro-authored. The ratified exception names the operator-messaging executor seat, a pinned build of the open-source Hermes agent runtime, as a permanent and disclosed component. Cerebro's ownership claim applies to the control plane around that seat: seam logic, allowlists, quarantine, credential scoping, per-seat contracts, budget keys, memory fanout and dispatch observability. The exception is narrow and does not turn other wrapped runtimes into owned infrastructure.

Nor does self-sufficiency mean every computation is local: locally hosted models running on owned hardware carry the workhorse load, while cloud judgment remains an explicit, consented tier selected by routing and cost policy.

Routing across that hybrid is tiered the way an organization is. Frontier judgment is reserved for the hardest calls, a senior-advisor tier handles principled review, mid rungs carry orchestration and summarization, and small high-throughput specialists do the volume work. Cost caps, latency, data placement and availability shift work between the tiers, and the tiers are enforced in the routing ledger as named rungs bound to explicit models, not described in prose alone.

Plane	Placement	Role
Operator edge	Operator workstation and mobile interfaces	Authoring, review, final sign-off and relationship actions
Always-on services	Always-on services node	Model gateway, vector store, scheduled jobs, Telegram seams and health probes

Plane	Placement	Role
GPU execution	Intermittent GPU node	Intermittent specialist inference, reranking, training and held-out scoring
Fallback/retained	Additional rostered hardware	Reversible fallback or future role; not assumed live without a probe

Figure 2. Functional placement by role, not a live-health claim and not a hardware roster. Node identity, availability and serving residency belong to the fleet ledger and a current probe.

The functional topology separates an operator edge, an always-on services plane and intermittent GPU execution. The services plane hosts the gateway, vector services, schedules and the Telegram seam; the GPU plane performs heavy inference, reranking, training and held-out scoring when available. This is placement, not a blanket live-health claim. The 30 July runtime ledger observed the Composer Telegram seam operating while other Constellation modules were code-present and not running. During the refreshed interval the model gateway's route table was reconciled against live state with a repository mirror, and its cost map was pinned into the service image, removing a cold-boot fetch from an external code host - a small but real reduction of the boot-time dependency surface.

8.1 Position-based runtime

The source-built Constellation architecture contains named positions and modules, but observed operation is narrower than the source tree. The recorded live seam uses Composer with the disclosed executor component. Broader Forge/Composer execution, durable effect semantics and additional module runtimes remain incomplete, frozen or parked. Profiles separate identity and configuration; they are not equivalent to OS-level security isolation. The source tree is broader than the three anchor positions: further named modules exist for tool execution and task lifecycle, context compression, output refinement and contained web probing, each ported from a studied pattern donor under the absorption discipline, and each labeled with its own operating status rather than presumed live.

Three named positions anchor the runtime's division of labor. Composer owns orchestration: context assembly, seams, session state and the observed operator-messaging path. Sentinel owns verification and evidence: the lane registry, promotion and pre-commit gates, receipt reconciliation and the close-to-open session seam. Scholar owns contained research: a sandboxed light/heavy retrieval-and-synthesis pipeline behind a fail-closed web gate, an isolated egress proxy and a secret-free fetch environment, with a reading agenda fed by Sentinel findings. A dated July 2026 internal benchmark compared Scholar against five published research-agent designs and recorded a stronger containment posture and a weaker iteration loop - a deliberate trade, since the containment is the part none of the surveyed systems carry. Composer's Telegram seam is the recorded operating path; Scholar and the remaining positions are source-built and probe-qualified rather than claimed live.

A Lane-B admission prototype was internally sabotage-tested for identity, bundle, manifest, host, freshness, nonce, approval and signature refusal behavior. It is not invoked by a production deployment path, and real seat unfreeze remains operator-gated. Likewise, the estate does not yet provide platform-grade swarm isolation or network-egress containment for an adversarial model with tools.

8.2 Command Center and the model factory

The Command Center remains a localhost-only, on-demand cockpit over fleet, memory, kanban, records, automations, specialist seats and the model factory. The model factory - Cerebro's training and evaluation pipeline, described in Section 9 - can produce and evaluate candidates, but several execution surfaces are parked and promotion remains a separate human act. Interface presence does not upgrade a dormant executor or an unbound model into operating capability.

Localhost is not a security boundary. Guarded browser actions, content-security policy and origin controls reduce accidental or policy-violating behavior, but known gaps remain between model-name policy and namespaced served aliases. This edition therefore treats origin policy as avoidance and provenance evidence, not as hostile-model containment.

8.3 Seats move as the frontier moves

The estate's model-agnosticism is mechanical, not rhetorical. Named routing entries bind roles to models, and each carries its serving route and a probe command; the routing ledger's version chain records each recorded rebinding as a dated note carrying its ratification pointer and revert recipe. Rollback fields are not yet uniform across the route contract itself; the revert paths live in the dated version notes and in repository history. The mechanism has been exercised: when re-benching favored newer admissible models, the local composer seat moved from an 8-billion-parameter Llama 3.1 build to Gemma 4 12B, and the mid-tier resolver moved from Phi-4 14B to Gemma 4 26B, with the retired routes verified dead rather than left dangling. Candidate models enter custody under recorded license and usage-policy review before any seat may resolve to them; custody precedes and never implies deployment. Which models are resident at any moment remains the fleet ledger's fact, established by a current probe rather than by this paper. What this paper claims is the contract machinery that lets the parts change without the governance changing.

The same posture governs frameworks. Current agent runtimes, harnesses and orchestration stacks are absorbed through a governed intake ladder: an immutable commit pin, a license gate, review of the code paths that would run hot, functional isolation and a recorded disposition. The estate table's most recent full reconciliation, dated inside this evidence window, rows all ninety-three top-level estate entries and records evidenced intake and native-implementation state; pin, license, review, caveat and extraction gaps remain explicit. Dated absorption decisions cover agent stacks, orchestration frameworks, observability integrations, skills donors and training-method research, including one method extraction whose donor weights were rejected while the method was retained. Patterns are adopted; runtimes are not surrendered.

8.4 The model estate at the evidence freeze

Breadth illustrates the estate's model range; the replaceability claim itself rests on the route contracts and recorded seat swaps of the preceding section. This edition names the breadth anyway, as a dated snapshot rather than a standing promise. The table below records the estate's model roster at the latest documented residency probe before the freeze, observed 12 August 2026. The live roster belongs to the fleet and routing ledgers and a current probe; this table exists so a reader can compare the snapshot against that live truth rather than take either on faith.

Role	Model at this freeze	Status at the latest recorded
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		probe
Operator messaging seam	Gemma 4 12B primary, with a Gemma 4 E4B fallback on the services plane	Observed operating
Trained finding-triage specialist	sentinel-scanner-cyber, the datar12 adapter, served at full precision	Promoted and serving
Judge of record at the transfer gate	olmo-3-judge 32B, digest-pinned; its base, OLMo 3.1 32B Instruct, is also resident	Ratified substrate
Senior-advisor and heavy-research tier	Gemma 4 31B	Seat-bound, resident
Mid-tier resolver	Gemma 4 26B	Seat-bound, resident
Research substrate	Nemotron 30B-A3B	Seat-bound, resident
Small specialist training base	Nemotron 3 Nano 4B	Resident
Embedding	nomic-embed-text	Resident on the GPU and services planes
Reranking	A dedicated cross-encoder behind the model gateway	Route-scoped service
Cloud judgment	Frontier routes under consent, routing and cost policy	Contracted tier
Custody candidates	Pulled with hash-verified receipts; several under open license or usage-policy review	Held un-served
Rejected and superseded artifacts	Prior adapters and runs retained content-addressed	On disk, unbound

At that probe, thirteen model tags were resident on the GPU plane and three on the services plane. The snapshot names the roles this paper claims and leaves current residency to the probe that owns it.

The breadth has history. A May 2026 championship bench ran seventeen candidate models, three runs per probe, against a fixed orchestration prompt, scoring tool obedience, coverage, honesty, latency and hallucination, and deliberately produced a nine-role specialist roster rather than a single winner. Candidates that fabricated under pressure were repurposed as council adversaries rather than discarded; a model with a strong refusal profile was assigned a safety role; and the bench record, its suppressions and its supersession path were preserved in the

decision log. Parts of that roster have since been superseded by the dated seat swaps recorded above, which is the point: the bench is an instrument the estate re-runs as the frontier moves, not a ranking it defends.

9. The model factory and the Sentinel evidence line

Cerebro's model factory trains and evaluates bounded adapters - small, task-specific trained model layers - on owned hardware, gating each candidate against held-out inference and evidence binding. Promotion itself is a recorded operator decision on that gate's evidence rather than an automatic consequence of a run, as the served adapter's disclosed exception below shows. At the evidence freeze, only one lane is proven and operating internally: cybersecurity finding triage for Sentinel. The factory's evaluation refinery and the Layer-0 arc demonstrate reusable candidate-production and comparison methods, not general model-factory maturity. During the interval three further origin-admissible model candidates were taken into custody under the same discipline: pulled to owned hardware, under recorded license and usage-policy review, and held un-served, so that acquisition never implies deployment.

The served artifact is Cerebro's own: sentinel-scanner-cyber, the datar12 adapter, trained on owned hardware from Cerebro-curated corpora. Its promotion is disclosed exactly as it happened. datar12 passed the ratified raw-judgment bar on all eight held-out suites under the gate's own inference path, but it retained two adversarial misses, so the strict zero-miss gate did not go green. It was promoted by a recorded operator decision under a codified dominance rule: both misses were already present among the thirty-five carried by the incumbent it replaced, and the served-path score on the 34-case promotion key, 0.88 against a 0.90 bar, is logged in the promotion record rather than absorbed into a pass. The exception is a named rule with its own decision record, and the zero-miss bar remains the standard against a clean incumbent.

Four training lanes are registered as executable contracts: cybersecurity triage, general-agent behavior, composer voice and research-claims grading. Each contract declares its recipe and held-out evaluations, and either carries its gate thresholds directly or points to the transfer gate that governs it. Two lanes have produced trained artifacts; only cyber triage is proven and serving, and a registered contract is construction, not a capability claim.

9.1 Gate first, adapter second

The trust boundary requires the gate to drive inference on held-out inputs, re-hash the adapter it loads and bind cases, outputs and evidence. Model provenance is recorded in a Model BOM (bill of materials). Promotion separates raw judgment, policy-derived outputs, adversarial regressions, statistical comparison and serving fidelity. A content-addressed winner can be retained without any consumer being redirected to it.

- Held-out inputs are digest-bound and separate from producer output.
- The scoring path owns inference and records the adapter actually loaded.
- A no-regression gate can block a model that improves one metric by breaking an adversarial or safety-critical case.
- Serving packaging is checked against the source adapter so quantization or runtime drift cannot inherit an unearned promotion.
- GPU launch and scoring use one serialized resource boundary rather than overlapping jobs that make evidence ambiguous.

The grader is as governed as the gate. Verdicts at the general-agent transfer gate are graded by a ratified judge of record: an independently developed, fully open model (OLMo 3.1 32B Instruct, Apache 2.0), digest-pinned, run on owned hardware, deliberately chosen from a different model family than the student it grades, and fail-closed when unavailable, with a margin-aware confirmation protocol for borderline verdicts and a disclosed break-glass path. Which judge grades which gate is recorded in a single canonical matrix, so a judge ratified for one gate cannot silently migrate to another. Cerebro did not train its own judge, on purpose: a maker grading its own family of students would narrow the very independence the gate exists to protect.

The judge was not a default; it was a bake-off with casualties, and the losing reasons are part of the record. A frontier route was withdrawn because its pinned version had gone stale. A hosted large-model route was ruled out because its free tier could not be pinned to an exact model and its data-handling terms could have exposed the held-out set. One local candidate, Command R, won a ten-for-ten sealed hard-probe veto and was still ruled out because its weights carry a non-commercial license; the weights were deleted from the estate rather than kept as temptation. The ratified judge won on the compound criterion that matters for a trust root: local, cross-family, digest-pinnable, reproducible, zero marginal cost, zero contamination exposure and permissively licensed. A previously ratified local fallback was later retired and its weights removed, so the gate now fails closed with a single disclosed break-glass path. A second, separate promotion gate uses a different ratified judge in paired-relative mode only, barred from absolute-threshold verdicts, and a single canonical matrix records which judge grades which gate so that neither ratification can silently migrate to the other.

Training method is inherited and then measured, not improvised. Lane recipes are grounded in the base model vendor's published fine-tuning guidance; where the small-data regime forced a deviation, the deviation and its size are documented in the canonical recipe rather than left as folklore. The training stack itself is open source.

9.2 What the datar and Layer-0 arcs proved

The cyber-triage arc ended with the best defensible serving decision rather than the highest run number. datar12 remained served after later rebalances moved errors among adjacent adversarial cases. A later run fixed targeted misses but retained an anchor regression and introduced new in-family misses, so the gate blocked it and the arc closed under a pre-declared stop condition.

The arc behind those endpoints is deeper than its endpoints. At this freeze the cyber lane's exam chain holds 582 held-out cases across eight standing suites, spanning adversarial transfer probes, blind instruction-injection resistance, robustness contrasts, a benign-negative flood, mined threat and host-posture sets and a merged control-mapping evaluation, plus a separate 34-case promotion key. Adapter-bound gate runs are designed to append to a score ledger kept beside the adapters, with development runs excluded so they cannot pollute the baseline history; an append failure is non-fatal and is printed rather than hidden, and serving decisions cite the available ledger rows and per-run artifacts rather than a favorite number. The cyber lane's corpus composition held 4,804 rows at its last recorded check; the served datar12 adapter was trained on its own separately pinned 4,599-row union. Most of the corpus is mined from real sources: open detection-query packs, host-configuration benchmark content, typed prompt-injection corpora and an authoritative public control-statement catalog, with hand-built generators supplying the targeted contrast the mined data lacked. Disjointness is enforced

mechanically, with exact sweeps against every held-out suite and targeted near-duplicate sweeps where that detector is valid; one evaluation-only corpus was deliberately never mined for training. The run history is equally candid: the arc iterated through named adapter runs and data-lever families, a preference-optimization attempt that washed out, and a reinforcement-learning sweep whose seven variants all scored flat and were recorded as exhausted rather than retried until one went green.

The factory also keeps a ledger of what it has not yet used. An extraction ledger tracks registered model-factory resources across acquisition, extraction and production, and it was born from a real failure: a five-thousand-question dataset sat pulled on disk while training ran on a few hundred generated rows, because nothing tracked extraction completeness. The rule since is that a resource is not done when pulled; it is done when extracted and in production, and the ledger's recorded pending list is an explicit factory backlog rather than a note buried in session prose.

The separate Layer-0 arc established a different boundary. A text-only, quantized served artifact outperformed its documented base arm on an internal 136-case comparison (121/136 vs 104/136, McNemar $p=0.00297$), was stored with content-addressed provenance and Model BOM evidence, and remained unbound. The result proves a packaging-and-comparison path over that bounded text set. It does not prove multimodal fidelity, external performance, a second production lane or a consumer deployment.

BOUNDED CLAIM / Cyber triage remains the only proven operating specialist lane. The text-only Layer-0 artifact passed a bounded internal served-path comparison and is content-addressed, but no consumer resolves to it. Neither result is external validation or general model-factory maturity.

10. Loop engineering across the workspace

Loop	Name	Cerebro-native meaning	Evidence status
4	Hill-climbing	Trace analysis proposes a bounded change to a Loop-1 prompt, skill or configuration	Open/proposal-only; never self-applies
3	Event-driven	Cron, launchd, Telegram messages, hooks and heartbeats invoke work without a manual start	Built on multiple lanes
2	Verification	Tests, evals, critics, provenance and trajectory checks decide ship or stop	Built; fourteen registered lanes at the v2.2 freeze, nine blocking and five advisory
1	Agent	A model or deterministic worker reads, calls tools and produces an artifact	Built across multiple surfaces

Loop engineering remains a useful vocabulary for triggers, checks and feedback, but the earlier advisor/council/orchestrator routing triad was retired after producing no real council runs. Its heuristic survived as doctrine; conscious invocation and ordinary adversarial review replaced a ritual that existed mostly as code and intent. Local seat routing remains a separate operational surface.

Figure 3. The four-loop lens. Loops 1-3 are operating patterns; Loop 4 remains gated proposal generation. Loop 2 enforcement status is registry-owned and re-checked per run.

Loop 1 exists across bounded workers and skills.

Loop 2 is the strongest documented pattern: tests, evals, critics, provenance and receipt checks decide whether selected outputs advance.

Loop 3 exists on scheduled and event-driven lanes, including the Telegram seam and health jobs. The presence of these patterns does not establish a general autonomous executor or a uniformly invoked orchestration platform.

Loop 4 remains proposal-only. Trace analysis may propose a bounded patch to a named prompt, skill or configuration, but the proposal re-enters independent verification and human approval before changing behavior. The return arrow is therefore governed rather than self-rewriting.

10.1 Existing loop families

Loop family	Trigger	Verifier	Return or stop condition
Memory health	Embed-on-write + daily recall checks	Coverage, tier, golden recall and scoped-route liveness	Repair proposal or deployment block
Model factory	Corpus or registered training run	Held-out eval, no-regression and serving fidelity	Promote, drop or close the arc
Commercial delivery	Opportunity intake and packaging workflow	Claim and evidence review + human approval gate	Draft package or stop
Content	Publishing cadence and production queue	Brand, claim and footage-safety review	The operator publishes or rejects
Repository hygiene	Pre-commit, CI and scheduled audits	Tests, path policy, provenance and drift detectors	Commit allowed, advisory filed or block

The consolidated loop-control code is Built, but operating invocation remains probe-qualified. The durable claim is narrower: the workspace contains multiple loop families and a shared vocabulary, while broad executor bodies, automatic self-improvement and fleet-wide orchestration are not established operating capabilities.

11. Security, provenance and recovery boundaries

Security in Cerebro is a set of scoped boundaries, not one universal wall. Credential guards primarily reduce accidental exposure by Cerebro's own agents. Path policy governs publication, memory, training and backup decisions. Model BOMs bind provenance. Profiles and allowlists constrain named seats. These controls support repository and operator assurance; they do not amount to certification, hostile-model containment or platform-grade swarm isolation.

During the refreshed interval the accidental-exposure guard was rebuilt, replacing two regex checks with a read-verb argument analyzer and adding fail-closed handling for split-string and environment-flag evasions. It was adversarially re-gated over five rounds - four driven by an external reviewer, the fifth an internal battery independently ratified by the decision layer - after which the residual findings in that family were closed and the rebuild arc was frozen. The guard remains an accidental-exposure control for trusted workers, not an adversarial containment boundary.

11.1 Path-based data classification

The merged path policy classifies repository locations without inspecting their contents and returns allow, review or deny for public extraction, memory ingestion, training and backup. An unmatched path fails closed to a conservative posture. A review label stops automation for human adjudication; it does not assert a formal legal or regulatory classification.

The policy is a shared decision engine, not an override of stricter controls. Its value is consistency across publication, corpus, memory and backup workflows. Its scope is also explicit: a repository path guard does not provide OS sandboxing, network-egress control or protection against an adversarial model that already has tools.

11.2 Model provenance and origin

The Model BOM and evidence gates bind trained or served artifacts to base ownership, source provenance, adapter evidence and route policy. Origin policy is an avoidance and approval control. A verified August gap register found that namespaced local aliases could bypass bare-name deny patterns. Until the owning control is repaired and re-probed, this paper does not claim a reliable hostile-origin lockout. Model estate hygiene continued in the same spirit during the refreshed interval: a resident model was pruned only after its manifest row was recorded, and an upstream license mislabeled as permissive was corrected to the vendor's own terms, keeping the license column a first-class recommendation-time fact.

11.3 Recovery: partial by design until proven

The nightly hub-store path creates consistent SQLite archives and, when present, a non-empty database dump; it rejects unsafe archives, checks SQLite integrity and writes a hash-bound marker. A separate 26 July status artifact records a verified off-host snapshot capture for thirteen Qdrant collections. These are meaningful capture proofs. They are not restore proofs or service-recovery proofs.

The recovery runbook proves SQLite artifact verification and limited scratch checks. The database dump has not been restored through an isolated service round trip, Qdrant snapshots have not been restored and queried in a timed drill, and model artifacts, configuration, credential rebinding and an encrypted second copy remain incomplete. Recovery readiness is therefore still partial.

12. The company operating system: Master Brain to departments

Department	Evidence-freeze state	Boundary
Ops / Master Brain	Active	Company governance, runtime, memory, security, shared skills and evidence
Business Development	Active	Pursuits, capability evidence and drafts; the operator owns external contact and commitment
Products	Active	Libro public alpha and BlackBox pre-ship program
Content	Active	Public distribution lane; the operator owns publication and engagement
Finance	Dormant-in-fact	Tracked figures are stale without dated owner input; no guessed burn or revenue
Cyber Services	Dormant	Offer materials exist; no active client or outreach authorization recorded
Training	Materials ready, pipeline dormant	Deck and bundles exist; no cohort, price or delivery commitment recorded

The repository remains a single-context company system with a deliberate chain of command. Master Brain owns company-wide governance, shared runtime contracts and cross-department evidence. Department brains own lane-specific sessions, decisions, pipeline state, drafts and delivery detail. Generated maps and inventories provide views; dated decisions and owning state pages remain the authority.

The generated inventory stamped 14 August 2026 records 470 decision documents, 71 company-wide skills, ten adopted principles, four active departments, three dormant departments and three active fleet nodes. Those are snapshot counts, not marketing claims. More important is the reproducible information architecture and the new canonical-skill parity: eleven shared runtime skills now resolve to one source, with an explicit warning that the primary checkout is the deployment surface.

The operating model remains Ops -> Products -> Content -> Consultations, but the August treadmill audit showed twelve consecutive infrastructure sessions while product lanes did not move. The ratified correction is not another scheduler: verification work must protect a named bid, model, sellable artifact or live operator capability. The next operating test is throughput into evidence-bearing product work.

13. Products, distribution and commercial state

13.1 Libro and BlackBox

Libro remains a public v0.2.1-alpha pre-release in a separate clean repository. A stable release remains gated on a fresh-customer experience check. BlackBox remains pre-ship.

Pricing, legal terms, external commitments and release approval remain human-owned. Nothing in the post-v2.1 commit volume upgrades either product's market state.

13.2 Business development and evidence transfer

Business development contains structured pursuit folders, capability libraries, evidence exhibits and owner-gated workflows. These are pipeline and capability facts. They are not proof of a repeatable deployment, an externally validated outcome or recurring revenue, and this public edition does not expose pursuit-specific material.

The strongest transfer from Ops to delivery is evidence discipline: deterministic demonstrations, serving receipts, sealed exhibits, claim classes and adversarial review records. Automation stops before external delivery, signature, representation, partnership or commitment. Internal proof must still be reconstructed in a customer-relevant environment before it becomes market proof.

13.3 Content and consultations

Content remains a distribution capability, and consultations remain downstream of product and evidence. No material post-v2.1 repository evidence upgraded these lanes into externally validated delivery. Publication and engagement remain operator-owned.

COMMERCIAL HONESTY: Artifacts are not outcomes. A draft is not a delivered package, a delivered package is not a win, a public alpha is not a stable product, service collateral is not an active client, and training materials are not a delivered cohort.

14. Conclusion

As of 14 August 2026, Cerebro is a real, internally exercised architecture and verification estate - not yet a repeatable externally validated platform.

The strongest evidence is specific: a receipt-oriented repository spine, operational memory with measured internal recall, one proven specialist lane served by a model Cerebro trained itself, a content-addressed but unbound general-agent artifact, transfer-gate promotion verdicts graded by a ratified independent judge, and one observed operator seam running on a pinned, disclosed open-source harness. The estate has also demonstrated the ability to catch false health, reject regressive models and narrow its own claims.

The principal gaps are equally specific: broader end-to-end execution, adversarial containment, service-level recovery, customer-reconstructable deployment and external outcome evidence. Cerebro remains architecturally ahead of its product surface. The fastest path to value is not more apparatus; it is converting one bounded capability into a repeatable, independently accepted result without weakening the evidence boundary.

This document is reversible. A later edition should name what changed, cite the successor evidence and preserve this version. A living white paper is credible only when a once-correct claim can expire and when internal proof is never silently promoted into market proof.

Appendix A. Claims and how to verify them

Each load-bearing claim in this paper binds to a repository artifact an inspecting party can be shown under the governance rules described in Section 12. Live runtime facts additionally require a current probe.

Claim	Evidence artifact	What an inspecting party is shown
Receipt ledgers are tamper-evident under an honest-operator model	Hash-chained per-repository receipt sinks	Chain re-verification over the ledger; an edited entry breaks the chain at the edited index, demonstrated end to end on a fresh clone of the public slice
Independent verification lanes can stop material transitions	Verification lane registry (fourteen lanes at this freeze: nine blocking, five advisory)	Show a blocking lane red stopping the installed Sentinel pre-commit gate; separately run <code>merge_when_green.py --selftest</code> to show a remote R7/check red stops the sanctioned merge client
Memory recall is measured and degradation is visible	Dated recall run receipts (recall@5 of 0.957 on 3 August 2026)	Run the offline fail-visible vector-lane test and inspect the dated 0.957 receipt separately; reproducing the live shard requires the private services and scoped credentials used by that run
The recall corpus is a designed federation, not one database	The memory architecture standard operating procedure and warehouse configuration	The eight designed collection classes and per-source write topology; live collection counts come from an authenticated probe
Cerebro trained the specialist it serves	The served model tag, the promoted pointer, the datar12 training-run manifest (factory-run-cyber-triage-20260704-0152.json), the promotion cutover record and the arc-closeout decision	The recorded serving tag and probe; the promoted pointer's adapter digests; the run manifest's base key, corpus digest and recipe, with the strict gate honestly recorded as not green and promotion not reached in-run; and the cutover and closeout records carrying the ratified

		dominance-rule exception, the raw-judgment scores and the revert path. A fresh live serving probe requires access to the private route
The general-agent student beat its base on a bounded set	The served-differential verdict, paired outputs and judge ballots, and the Layer-0 Model BOM (121/136 versus 104/136, McNemar $p=0.00297$)	The 136 per-task verdict entries and bound output digests sufficient to recount the two arms and recompute the paired comparison; the artifact remains bound to no consumer
Transfer-gate promotion verdicts are independently graded	The judge ratification matrix and the fail-closed transfer-gate evaluator	The matrix row naming the digest-pinned judge of record, its ratification record, reopen bar and enforcement point
The model estate snapshot is dated and probe-backed	The fleet and routing ledgers and the latest recorded residency probe	The dated roster snapshot beside the live ledger and a current probe, so the snapshot is re-taken rather than trusted
Held-out evaluation is broad, not anecdotal	The eight standing exam suites and the separate promotion key (582 plus 34 cases at this freeze)	The suite case files with hash-frozen input variants and the disjointness sweep results
Promotion-grade score history is recorded	The adapter-side score ledger and committed validation snapshots	Available rows binding adapters, suites and dates, with append limits and exclusions stated rather than hidden
Pulled is not done	The extraction ledger and its status ladder	Rows for registered resources across acquisition, extraction and production, including the recorded pending list; comparison with the dated runtime inventory surfaces unregistered pulls
Seats bind to models through named, versioned routing entries	The model route contract and the routing ledger's version chain	A named route entry with its probe command; dated ledger version notes recording past seat swaps with ratification

		pointers and revert recipes
Current frameworks are absorbed without surrendering the runtime	The vendored-imports estate table and dated absorption decisions	Inventory coverage, evidenced pin, license and review fields where present, intake and native state, operator disposition, and explicit recheck gaps
Backups are captured and verified; end-to-end service recovery remains unproven	Hash-bound backup markers and the dated snapshot status artifact	Capture and integrity evidence, the dated record of the bounded Qdrant collection restore drill, and the remaining timed multi-service restore gaps
The installable slice is public	The Libro public repository	The tagged v0.2.1-alpha pre-release in a separate clean repository
Major capability claims are bounded by a status vocabulary an audit can apply	This paper's status vocabulary and the evidence-cut review notes	Compare the vocabulary against the abstract, Sections 8-14 and Appendix A; this is a manual editorial review rule, not per-claim machine tagging

Appendix B. Selected references

Internal Cerebro publications are listed first, followed by the external methods, software and research designs referenced in this paper. Internal documents are available for inspection under the governance rules described in Section 12.

- Cerebro Cyber Solutions. (2026, May 28). Cerebro: Operator-owned AI infrastructure as a governance discipline (Version 1.3.3, public edition).
- Cerebro Cyber Solutions. (2026). Supplement 1: Day 45 to Day 65.
- Cerebro Cyber Solutions. (2026, April 20). The personal AI operating system: Five-layer doctrine.
- Cerebro Cyber Solutions. (2026, April 21). Index, harness, orchestration: Three-layer doctrine.
- Cerebro Cyber Solutions. (2026, June 25). Cerebro Principle #10: Verified completion.
- Cerebro Cyber Solutions. (2026, June 23). Loop engineering: The four-loop framework, mapped to Cerebro.
- Cerebro Cyber Solutions. (2026, July 9). Memory architecture: Standard operating procedure (last verified edition).
- Cerebro Cyber Solutions. (2026, July 8). datar arc closeout: Serve datar12.
- Cerebro Cyber Solutions. (2026). Data classification and path-routing policy.

- Cerebro Cyber Solutions. (2026, August 5). Workspace bootstrap and hub store recovery (verified edition).
- Cormack, G. V., Clarke, C. L. A., & Buettcher, S. (2009). Reciprocal rank fusion outperforms Condorcet and individual rank learning methods. In Proceedings of SIGIR '09 (pp. 758-759). ACM.
- McNemar, Q. (1947). Note on the sampling error of the difference between correlated proportions or percentages. Psychometrika, 12(2), 153-157.
- Nogueira, R., & Cho, K. (2019). Passage re-ranking with BERT (arXiv:1901.04085). arXiv.
- AIDE: AI-driven exploration in the space of code (arXiv:2502.13138). (2025). arXiv.
- Deep research agents: A systematic examination and roadmap (arXiv:2506.12594). (2025). arXiv.
- Stanford Open Virtual Assistant Lab. (2024). STORM: Synthesis of topic outlines through retrieval and multi-perspective question asking [Computer software]. GitHub. <https://github.com/stanford-oval/storm>
- Elovic, A. (2026). GPT Researcher [Computer software]. GitHub. <https://github.com/assafelovic/gpt-researcher>
- UncleCode. (2026). Crawl4AI [Computer software]. GitHub. <https://github.com/unclecode/crawl4ai>
- NVIDIA. (2026). Nemotron usage cookbook [Computer software]. GitHub. <https://github.com/NVIDIA-NeMo/Nemotron>
- Google. (2026). Gemma cookbook [Computer software]. GitHub. <https://github.com/google-gemma/cookbook>
- Unsloth AI. (2026). Unsloth [Computer software]. GitHub. <https://github.com/unslothai/unsloth>
- Qdrant. (2026). Qdrant vector database [Computer software]. <https://qdrant.tech>